LEARNER GUIDE

A table can predict a line

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A straight-line rule builds a table one row at a time — and just a few rows already contain the whole line. In this lesson you use a rule to work out the y value for each x, notice that the y values go up (or down) by the same amount every step, and use that steady step to predict the line before you ever draw it. You also test whether a pattern really is a straight line by checking that its step stays constant. The question you are exploring is: how can a few coordinate pairs reveal a whole linear pattern?

What you are learning to notice

- That you can generate coordinates from a rule by substituting each x.
- That in a straight-line table the y values change by the same amount each step — the first difference.
- That this constant step lets you predict the line's direction and any future value.
- That a pattern is only a straight line if its step stays constant — not every rule is linear.

Moving through it, one step at a time

In the Table-to-Line Predictor, use the rule to fill each y, check, and notice the constant step — then predict which way the line goes before revealing the graph. In Step-to-Build, press '+ step' to grow the table one row at a time and watch the line build. In First-Difference Detective, decide whether each table's step is constant. Then predict a mystery line from just its first two rows. There is no rush; use the coordinate grid on the Scratchpad to plot and predict.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (substituting a value into a rule such as $y = 3x + 2$, and reading and plotting coordinates), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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